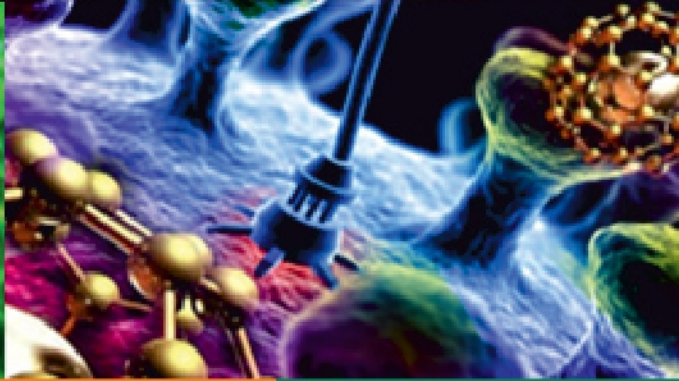
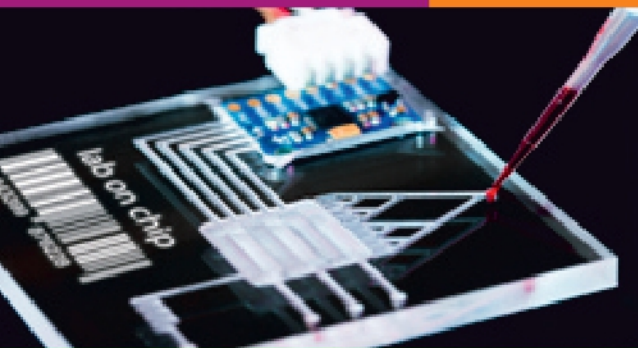


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(ICRTB 2024)

January 19-20, 2024

**ABSTRACTS**



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**PP-039**

**Molecular docking and simulation studies of Curcumin derivatives against COX-2 in Alzheimer's Disease**

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**Abstract:** Natural compounds, including those from medicinal plants, show promise in healthcare, addressing neurological and cardiovascular disorders. The use of NSAIDs in neurological conditions raises concerns about cardiovascular risks, emphasizing the need for careful consideration. Drug repurposing, particularly with curcumin derivatives, offers a potential avenue for addressing these challenges in drug development. Its potential therapeutic benefits extend to conditions like Alzheimer's disease by influencing immune responses and mitigating oxidative stress. In our research work, curcumin derivatives with 90% similarity were extracted from the PubChem database, resulting in 2561 compounds. The Cyclooxygenase -2 is considered a target protein for docking studies against curcumin derivatives. The inhibitor site was noted in the crystal structure of cyclooxygenase-2 (PDBID: 6COX) and the grid box was determined for site site-directed docking procedure. The docking of the 2561 ligands was executed through Perl programming and AutoDock Vina, generating output files with docking scores. Python code was then employed to extract top scores, stored in a text file, and transferred to Excel for analysis. Finally, PyMol facilitated the preparation of protein-ligand complexes, with the top 20 ligands analyzed and their best configurations saved for further examination. In a comprehensive screening of 2561 curcumin derivatives against COX-2 proteins, *in silico* docking revealed significantly higher scores compared to conventional COX-2 inhibitors. This suggests a strong binding affinity of curcumin derivatives to COX-2. Further steps involve synthesizing novel derivatives for molecular dynamics and subsequent *in vitro* assessments, including toxicity and cell viability, to assess their therapeutic potential.



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ISBN 978-93-6039-051-8

